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Top Skills

Data Science

LangChain

Amazon Web Services (AWS)

Manikanth Goud Gurujala

Data Scientist | RAG/LLM & Agentic AI Systems | ML Engineering |
MDS @ UBC '26

Vancouver, British Columbia, Canada

Summary

I am a data professional with a unique background in Mechatronic Engineering, dedicated to solving complex business challenges through advanced analytics and artificial intelligence. Currently based in Vancouver, I focus on building data-driven technologies that optimize operations and deliver strategic insights. What I do:-

- AI & Machine Learning: Developing predictive models utilising supervised machine learning, deep learning architectures, and Bayesian inference. I focus on hyperparameter optimisation, real-time computer vision pipelines, and deploying scalable models to cloud infrastructure (AWS EC2, FastAPI) to solve complex, real-world problems.
- Generative AI: Architecting scalable LLM and hybrid RAG pipelines (combining lexical and dense vector search) to automate workflows. I specialise in processing massive datasets efficiently out-of-memory using tools like DuckDB and Parquet to enhance information retrieval.
- Data Engineering & Analytics: Designing cloud-native ETL pipelines (Snowflake), open-source data validation frameworks distributed via PyPI, and establishing automated CI/CD testing workflows to ensure data reproducibility.
- Cross-Functional Collaboration: Translating complex statistical concepts into clear, commercial strategies. Drawing on my experience as a student liaison, I excel at building relationships and communicating technical insights effectively to diverse business stakeholders. I am always eager to tackle new challenges in the AI space and connect with fellow data professionals. Let's build something impactful together.

Experience

VRIFY

Data Scientist

April 2026 - June 2026 (3 months)

Vancouver, BC

Collaborating with VRIFY on a full time Capstone project "Deep Learning-Based Prediction of Soil Geochemistry Using Multi-Source Raster Data" as a part of UBC Master of Data Science Program

Built and deployed a multi-task CNN model to production for geochemical concentration prediction across mineral exploration sites, replacing VRIFY's existing baseline system.

- Architected end-to-end ML pipelines on Databricks processing multi-source geospatial datasets (geochemical assays, remote sensing, geophysical surveys), with automated feature engineering and data validation frameworks.
- Designed a multi-task CNN architecture that outperformed VRIFY's production baseline across all 10 geochemical prediction targets, validated using Stratified Spatial Cross-Validation to prevent spatial data leakage.
- Integrated SHAP explainability to make model predictions interpretable for geologists, enabling defensible decision-making on Areas of Interest (AOIs) in the field.
- Managed the full model lifecycle using MLflow model registry for experiment tracking, versioning, and reproducibility, with the final model package deployed to VRIFY's production environment.
- Collaborated with a cross-functional team of data scientists and domain experts to deliver a production-ready system within a 3-month timeline.

UltraTech Cement

AI/ML Engineer

October 2024 - December 2024 (3 months)

Mumbai

- Developed and deployed YOLOv8-based computer vision models for real-time safety monitoring, achieving over 88% detection accuracy across multiple industrial sites and improving incident response times by 30%.
- Collaborated with the Central Procurement Cell to analyse operational data and optimise fuel procurement and logistics using discrete event simulations, reducing transportation costs by 15% and improving scheduling efficiency.
- Utilised OpenAI's Generative AI frameworks to automate image and video summarisation, enhancing safety reporting efficiency and cutting manual review time by 40%.
- Developed dashboards to monitor costs and detect operational anomalies, improving resource allocation by 20%.

JUMP! Foundation

Global Facilitator
January 2024 - February 2024 (2 months)
Bangkok City, Thailand

Asia Pacific University of Technology and Innovation (APU / APIIT)
Research Assistant
May 2023 - October 2023 (6 months)
WP. Kuala Lumpur

- Architected and deployed an autonomous drone-based monitoring system for aeroponics farms, integrating custom CNN models to achieve 95% disease detection accuracy. This automation eliminated the need for manual scouting, significantly reducing labor costs and human error in crop management.
- Spearheaded multidisciplinary research and literature synthesis to establish the foundational framework for AI-driven precision agriculture. Collaborated across engineering teams to translate theoretical models into functional drone systems, accelerating the project's transition from conceptual research to real-world field testing.
- Initiated phase 3, which was the end-to-end integration of AI algorithms with UAV technology, conducting extensive field trials to optimise real-time data collection.

Orieus Electronics (M) Sdn Bhd
Automation and Robotics Intern
September 2022 - February 2023 (6 months)
Shah Alam, Selangor, Malaysia

- Proficiently programmed industrial and collaborative robots from FANUC, Nachi, Aubo, Omron, and other manufacturers such as Yamaha.
- Played a crucial role as a team member in a press shop automation project at Johnson Controls Hitachi, utilizing 5 collaborative robots.
- Executed PLC programming and HMI programming to effectively control and monitor automated systems.
- Assisted in a centralized packaging project at Panasonic, programming 17+ FANUC M-20iD/25 industrial robots.
- Successfully optimized the programming of robots, resulting in a 30% increase in production time.
- Demonstrated adaptability, strong technical skills, and a passion for automation and robotics.

Asia Pacific University of Technology and Innovation (APU / APIIT)
Engineering Technical Assistant
February 2022 - August 2022 (7 months)

Malaysia

Telefina Communication Sdn Bhd

Freelance Consultant

October 2021 - December 2021 (3 months)

WP. Kuala Lumpur

Provide support for survey report documentation

-Draft mini technical proposals

Churros n Co

Barista

June 2016 - October 2016 (5 months)

Kuala Lumpur, Federal Territory of Kuala Lumpur, Malaysia

GOPIO

Assistant

October 2015 - November 2015 (2 months)

Kuala Lumpur, Federal Territory of Kuala Lumpur, Malaysia

Ensure efficient administrative system. Organization in the office alongside responsibilities to ensure smooth running of various business conferences.

Education

The University of British Columbia

Master of Data Science · (August 2025 - June 2026)

De Montfort University

Master of Engineering - MEng, Mechatronics Engineering · (2024)

Asia Pacific University of Technology and Innovation (APU / APIIT)

Bachelor of Engineering - BEng, Mechatronics, Robotics, and Automation

Engineering · (2019 - 2023)

Brickfields Asia College

GCE A Level, Mathematics, Physics, Chemistry and Economics · (January

2016 - August 2017)